Multi-parameter water quality sonde

iLab 901

Up to seven water quality parameters at once, from one rugged sonde and its handheld.

The iLab 901 is built for long-term water quality monitoring. Fit it with up to seven sensors, chosen from thirteen measurements, and it cleans them itself so the data stays accurate for longer.

The handheld shows each reading on a 3.5 inch colour screen and stores more than 100,000 of them, with a GPS receiver and USB built in.



iLab 901 sonde with its sensor guard removed

7	13	100,000+	IP68	RS485
Parameters at once	Measurements to choose from	Readings stored	Sonde, to 30 m deep	Modbus output

What you get

01 Seven at once
One sonde carries physical, electrochemical, optical and ion-selective sensors, picked for the job.

02 Cleans itself
A self-cleaning system keeps the sensors clear, so the iLab holds accurate data for longer.

03 Long deployments
Set the sampling interval to suit, and stretch the time the sonde runs on its own lithium cells.

04 Readings in your hand
A 3.5 inch colour screen with an adjustable backlight shows results live, with fast processing on board.

05 Built for the field
An IP68 sonde and an IP67 handheld with high-impact rubber gaskets and a shape that is easy to hold.

06 Located and connected
GPS and USB are built in and Bluetooth is optional. The sonde speaks Modbus over RS485.

Where it works

Rivers, lakes and reservoirs
Online multi-parameter water quality monitoring.

Source water
Drinking water sources, groundwater and sea water.

Environmental monitoring
Long-term water quality records.

Agriculture
Water quality monitoring for agriculture.

Measurements

Thirteen measurements. Fit up to seven.

Each iLab 901 is fitted with the sensors your program needs. Conductivity also gives salinity and total dissolved solids.

Physical	2
Electrochemical	3
Optical	4
Ion-selective	4

Measurement	Method	Range	Resolution	Accuracy
Physical				
Temperature	Thermistor	0 to 60 °C	0.01 °C	±0.5 °C
Depth	Pressure	0 to 61 m	2 cm	±0.3%
Electrochemical				
pH	Glass electrode	0 to 14 pH	0.01 pH	±0.1 pH
ORP	Glass electrode	–1000 to +1000 mV	1 mV	±5 mV
Conductivity with salinity and TDS	Platinum gauze electrode pair	1 to 2000 µS/cm (K = 1) 100 µS/cm to 100 mS/cm (K = 10)	0.1 µS/cm to 0.01 mS/cm, by range	±3%
Optical				
Dissolved oxygen	Fluorescence	0 to 20 mg/L (0 to 20 ppm) 0 to 200%	0.01 mg/L 0.1%	±3% or ±0.3 mg/L, whichever is greater
Turbidity	Light scattering	0 to 1000 NTU	0.1 NTU	±5% or 0.3 NTU, whichever is greater
Chlorophyll	Fluorescence	0 to 500 µg/L	0.01 to 0.1 µg/L, by range	±5% of the signal level of 1 ppb rhodamine B dye
Blue-green algae	Fluorescence	200 to 300,000 cells/mL	20 cells/mL	±10% of the signal level of 1 ppb rhodamine B dye
Ion-selective				
Ammonia nitrogen	Ion-selective electrode	0.1 to 1000 mg/L-N	0.01 mg/L-N	±5% or ±0.2 mg/L
Nitrate nitrogen	Ion-selective electrode	0.5 to 1000 mg/L-N	0.01 mg/L-N	±10% or ±0.2 mg/L
Chloride	Ion-selective electrode	3 to 1000 mg/L	0.01 mg/L	±10% or ±0.2 mg/L
Fluoride	Ion-selective electrode	0.5 to 1000 mg/L	0.01 mg/L	±10% or ±0.2 mg/L

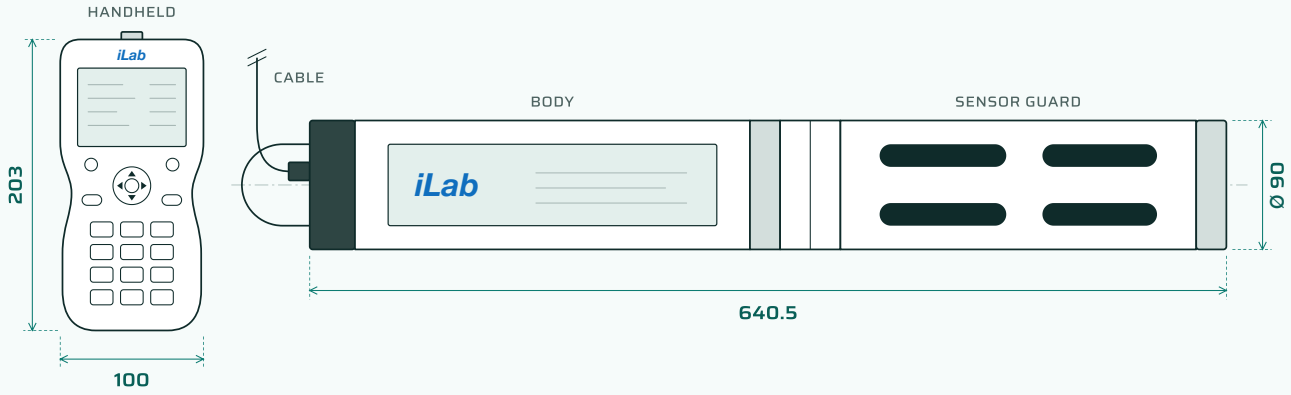
With an ion-selective sensor
The pH working range is 5 to 10 pH, and the sonde is rated to a depth of 10 m.

Depth
The depth sensor reads to 61 m. The sonde itself is rated to 30 m.

Temperature
The sonde measures in water from 0 to 40 °C, non-freezing.

Specifications

Sonde and handheld.



Outline drawing, both at the same scale. Dimensions in mm. The handheld is 43 mm deep.

Sonde

iLab 901

Sensors	Up to 7 at once
Communication	Modbus, RS485
Power supply	9 to 36 V DC
Power draw	3 W
Battery	8 × 3.6 V lithium cells
Protection	IP68
Depth rating	30 m (10 m with an ion-selective sensor)
Measuring temperature	0 to 40 °C, non-freezing
Storage temperature	0 to 50 °C, non-freezing
Size	90 mm diameter × 640.5 mm long
Weight	3 kg

Handheld

iLab communicator

Display	3.5 inch colour screen, adjustable backlight
Data storage	More than 100,000 readings
Positioning	GPS receiver, built in
Interfaces	USB; Bluetooth optional
Power supply	4 × 3.7 V rechargeable lithium, built in
Protection	IP67
Material	ABS and polycarbonate
Operating temperature	0 to 50 °C, non-freezing
Storage temperature	–15 to 60 °C
Size	203 × 100 × 43 mm
Weight	0.5 kg

Specifications are subject to change without notice.

EDS Environmental Data Services

Monitoring Australia's water, wastewater and environment since 1991.

Tell us what you need to measure.

Give us the water, the parameters and how long the sonde will be out. We'll match the sensors to your site and come back with pricing.

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